

6(a)(i)	haem (group) ; I Fe^{2+} I prosthetic group	1
6(a)(ii)	(iron in haem group) combines / binds, with (one) oxygen (molecule) ; A carries / transports / attaches, oxygen A 'bonds with' if stated as iron ion / Fe^{2+} / ferrous ion R oxygen atom / 2 or more oxygens R carbon dioxide binds	1
6(a)(iii)	more than one <u>polypeptide</u> (chain) ; A 2 or more R more than 2 / many / multiple A haemoglobin has four <u>polypeptides</u> I amino acid chain	1
6(b)(i)	accept values within the range 10–14 <u>kPa</u> ; A as a range or as a single figure	1
6(b)(ii)	accept any <u>range</u> between 1–6 <u>kPa</u> ; A within range 0–6 kPa R a single figure <i>ref. to</i> (steep) decrease in percentage saturation of haemoglobin as $p\text{O}_2$ decreases ; R increasing saturation A (oxy)haemoglobin, dissociates / releases its oxygen A haemoglobin has a low affinity for oxygen	2

Question	Answer	Marks
6(b)(iii)	<i>units – kPa and % – must each be used once</i> 1 (dissociation) curve shifts to the right ; I curve shifts downwards / graph if given for curve 2 (percentage) saturation (of haemoglobin) decreases (as $p\text{CO}_2$ increases) ; A decrease affinity of Hb for O_2 3 comparative data quote giving percentage saturation at the same $p\text{O}_2$ and at two different values of $p\text{CO}_2$; A percentage of oxygen released 4 <i>any ref. to</i> difference between position of curves, e.g. large difference in percentage saturation in middle of the range / little difference at high $p\text{O}_2$;	3
6(b)(iv)	Bohr, shift / effect ;	1
6(b)(v)	<i>any two from:</i> supplies more oxygen (to respiring tissues) ; A more oxygen is released / AW I quicker / faster I haemoglobin releases oxygen more readily <i>idea that</i> allows oxygen to be supplied to, (named) tissues / cells, to meet demand / as demand increases ; (maintains) aerobic respiration ;	2